

- JAHNKE, E., and EMDE, F. 1945. *Tables of Functions*, 4th ed., Dover, New York.
- JAIN, V. K., and CROCHIERE, R. E. 1984. "Quadrature Mirror Filter Design in the Time Domain," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-32, pp. 353-361, April.
- JENKINS, G. M., and WATTS, D. G. 1968. *Spectral Analysis and Its Applications*, Holden-Day, San Francisco.
- JOHNSTON, J. D. 1980. "A Filter Family Designed for Use in Quadrature Mirror Filter Banks," *IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 291-294, April.
- JOHNSON, D. H. 1982. "The Application of Spectral Estimation Methods to Bearing Estimation Problems," *Proc. IEEE*, Vol. 70, pp. 1018-1028, September.
- JONES, R. H. 1976. "Autoregression Order Selection," *Geophysics*, Vol. 41, pp. 771-773, August.
- JURY, E. I. 1964. *Theory and Applications of the z-Transform Method*, Wiley, New York.
- KAILATH, T. 1974. "A View of Three Decades of Linear Filter Theory," *IEEE Trans. Information Theory*, Vol. IT-20, pp. 146-181, March.
- KAILATH, T. 1981. *Lectures on Wiener and Kalman Filtering*, 2nd Printing, Springer-Verlag, New York.
- KAILATH, T. 1985. "Linear Estimation for Stationary and Near-Stationary Processes," in *Modern Signal Processing*, T. Kailath, ed., Hemisphere Publishing Corp., Washington, D.C.
- KAILATH, T. 1986. "A Theorem of I. Schür and Its Impact on Modern Signal Processing," in Gohberg (1986).
- KAILATH, T., VIEIRA, A. C. G., and MORF, M. 1978. "Inverses of Toeplitz Operators, Innovations, and Orthogonal Polynomials," *SIAM Rev.*, Vol. 20, pp. 1006-1019.
- KAISER, J. F. 1963. "Design Methods for Sampled Data Filters," *Proc. First Allerton Conference on Circuit System Theory*, pp. 221-236, November.
- KAISER, J. F. 1966. "Digital Filters," in *System Analysis by Digital Computer*, F. F. Kuo and J. F. Kaiser, Eds., Wiley, New York.
- KALOUPSIDIS, N., and THEODORIDIS, S. 1987. "Fast Adaptive Least-Squares Algorithms for Power Spectral Estimation," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-35, pp. 661-670, May.
- KALMAN, R. E. 1960. "A New Approach to Linear Filtering and Prediction Problems," *Trans. ASME, J. Basic Eng.*, Vol. 82D, pp. 35-45, March.
- KALMAN, R. E., and BUCY, R. S. 1961. "New Results in Linear Filtering Theory," *Trans. ASME, J. Basic Eng.*, Vol. 83, pp. 95-108.
- KASHYAP, R. L. 1980. "Inconsistency of the AIC Rule for Estimating the Order of Autoregressive Models," *IEEE Trans. Automatic Control*, Vol. AC-25, pp. 996-998, October.
- KAVER, J., and BRUZZONE, S. P. 1979. "Order Determination for Autoregressive Spectral Estimation," *Record of the 1979 RADCS Spectral Estimation Workshop*, pp. 139-145, Griffin Air Force Base, Rome, N.Y.
- KAVER, M., and LIPPERT, G. A. 1983. "An Optimum Tapered Burg Algorithm for Linear Prediction and Spectral Analysis," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-31, pp. 438-444, April.

- KAY, S. M. 1980. "A New ARMA Spectral Estimator," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-28, pp. 585-588, October.
- KAY, S. M. 1988. *Modern Spectral Estimation*, Prentice Hall, Englewood Cliffs, N. J.
- KAY, S. M., and MARPLE, S. L., JR. 1979. "Sources of and Remedies for Spectral Line Splitting in Autoregressive Spectrum Analysis," *Proc. 1979 ICASSP*, pp. 151-154.
- KAY, S. M., and MARPLE, S. L., JR. 1981. "Spectrum Analysis: A Modern Perspective," *Proc. IEEE*, Vol. 69, pp. 1380-1419, November.
- KESLER, S. B., ed. 1986. *Modern Spectrum Analysis II*, IEEE Press, New York.
- KNOWLES, J. B., and OLCAYTO, E. M. 1968. "Coefficient Accuracy and Digital Filter Response," *IEEE Trans. Circuit Theory*, Vol. CT-15, pp. 31-41, March.
- KRISHNA, H. 1988. "New Split Levinson, Schür, and Lattice Algorithms for Digital Signal Processing," *Proc. 1988 International Conference on Acoustics, Speech, and Signal Processing*, pp. 1640-1642, New York, April.
- KROMER, R. E. 1969. "Asymptotic Properties of the Autoregressive Spectral Estimator," Ph.D. dissertation, Department of Statistics, Stanford University, Stanford, Calif.
- KUNG, S. Y., and HU, Y. H. 1983. "A Highly Concurrent Algorithm and Pipelined Architecture for Solving Toeplitz Systems," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-31, pp. 66-76, January.
- KUNG, S. Y., WHITEHOUSE, H. J., and KAILATH, T., eds. 1985. *VLSI and Modern Signal Processing*, Prentice Hall, Englewood Cliffs, N.J.
- LACOSS, R. T. 1971. "Data Adaptive Spectral Analysis Methods," *Geophysics*, Vol. 36, pp. 661-675, August.
- LANG, S. W., and MCCLELLAN, J. H. 1980. "Frequency Estimation with Maximum Entropy Spectral Estimators," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-28, pp. 716-724, December.
- LEVINSON, N. 1947. "The Wiener RMS Error Criterion in Filter Design and Prediction," *J. Math. Phys.*, Vol. 25, pp. 261-278.
- LEVY, H., and LESSMAN, F. 1961. *Finite Difference Equations*, Macmillan, New York.
- LIU, B. 1971. "Effect of Finite Word Length on the Accuracy of Digital Filters—A Review," *IEEE Trans. Circuit Theory*, Vol. CT-18, pp. 670-677, November.
- MAKHOUL, J. 1975. "Linear Prediction: A Tutorial Review," *Proc. IEEE*, Vol. 63, pp. 561-580, April.
- MAKHOUL, J. 1978. "A Class of All-Zero Lattice Digital Filters: Properties and Applications," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-26, pp. 304-314, August.
- MARKE, J. D., and GRAY, A. H., JR. 1976. *Linear Prediction of Speech*, Springer-Verlag, New York.
- MARPLE, S. L., JR. 1980. "A New Autoregressive Spectrum Analysis Algorithm," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-28, pp. 441-454, August.
- MARPLE, S. L., JR. 1987. *Digital Spectral Analysis with Applications*, Prentice Hall, Englewood Cliffs, N.J.
- MARZETTA, T. L. 1983. "A New Interpretation for Capon's Maximum Likelihood Method of Frequency-Wavenumber Spectral Estimation," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-31, pp. 445-449, April.

- MARZETTA, T. L., and LANG, S. W. 1983. "New Interpretations for the MLM and DASE Spectral Estimators," *Proc. 1983 ICASSP*, pp. 844-846, Boston, April.
- MARZETTA, T. L., and LANG, S. W. 1984. "Power Spectral Density Bounds," *IEEE Trans. Information Theory*, Vol. IT-30, pp. 117-122, January.
- MASON, S. J., and ZIMMERMAN, H. J. 1960. *Electronic Circuits, Signals and Systems*, Wiley, New York.
- McCLELLAN, J. H. 1982. "Multidimensional Spectral Estimation," *Proc. IEEE*, Vol. 70, pp. 1029-1039, September.
- McDONOUGH, R. N. 1983. "Application of the Maximum-Likelihood Method and the Maximum Entropy Method to Array Processing," in *Nonlinear Methods of Spectral Analysis*, 2nd ed., S. Haykin, ed., Springer-Verlag, New York.
- McGILLEM, C. D., and COOPER, G. R. 1984. *Continuous and Discrete Signal and System Analysis*, 2nd ed., Holt Rinehart and Winston, New York.
- MEDITCH, J. E. 1969. *Stochastic Optimal Linear Estimation and Control*, McGraw-Hill, New York.
- MILLS, W. L., MULLIS, C. T., and ROBERTS, R. A. 1981. "Low Roundoff Noise and Normal Realizations of Fixed-Point IIR Digital Filters," *IEEE Trans. Acoustics, Speech, and Signal processing*, Vol. ASSP-29, pp. 893-903, August.
- MORF, M., VIEIRA, A., and LEE, D. T. 1977. "Ladder Forms for Identification and Speech Processing," *Proc. 1977 IEEE Conference Decision and Control*, pp. 1074-1078, New Orleans, La., December.
- MULLIS, C. T., and ROBERTS, R. A. 1976a. "Synthesis of Minimum Roundoff Noise Fixed-Point Digital Filters," *IEEE Trans. Circuits and Systems*, Vol. CAS-23, pp. 551-561, September.
- MULLIS, C. T., and ROBERTS, R. A. 1976b. "Roundoff Noise in Digital Filters: Frequency Transformations and Invariants," *IEEE Trans. Acoustics, Speech and Signal Processing*, Vol. ASSP-24, pp. 538-549, December.
- MUSICUS, B. 1985. "Fast MLM Power Spectrum Estimation from Uniformly Spaced Correlations," *IEEE Trans. Acoustics, Speech, and Signal Proc.*, Vol. ASSP-33, pp. 1333-1335, October.
- NEWMAN, W. I. 1981. "Extension to the Maximum Entropy Method III," *Proc. 1st ASSP Workshop on Spectral Estimation*, pp. 1.7.1-1.7.6, Hamilton, Ontario, Canada, August.
- NIKIAS, C. L., and RAGHUVEER, M. R. 1987. "Bispectrum Estimation: A Digital Signal Processing Framework," *Proc. IEEE*, Vol. 75, pp. 869-891, July.
- NIKIAS, C. L., and SCOTT, P. D. 1982. "Energy-Weighted Linear Predictive Spectral Estimation: A New Method Combining Robustness and High Resolution," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-30, pp. 287-292, April.
- NUTTALL, A. H. 1976. "Spectral Analysis of a Univariate Process with Bad Data Points, via Maximum Entropy and Linear Predictive Techniques," *NUSC Technical Report TR-5303*, New London, Conn., March.
- NYQUIST, H. 1928. "Certain Topics in Telegraph Transmission Theory," *Trans. AIEE*, Vol. 47, pp. 617-644, April.
- OPPENHEIM, A. V. 1978. *Applications of Digital Signal Processing*, Prentice Hall, Englewood Cliffs, N.J.

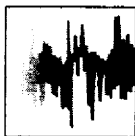
- OPPENHEIM, A. V., and SCHAFER, R. W. 1989. *Discrete-Time Signal Processing*, Prentice Hall, Englewood Cliffs, N.J.
- OPPENHEIM, A. V., and WEINSTEIN, C. W. 1972. "Effects of Finite Register Length in Digital Filters and the Fast Fourier Transform," *Proc. IEEE*, Vol. 60, pp. 957-976, August.
- OPPENHEIM, A. V., and WILLSKY, A. S. 1983. *Signals and Systems*, Prentice Hall, Englewood Cliffs, N.J.
- PAPOULIS, A. 1962 *The Fourier Integral and Its Applications*, McGraw-Hill, New York.
- PAPOULIS, A. 1984. *Probability, Random Variables, and Stochastic Processes*, 2nd ed., McGraw-Hill, New York.
- PARKER, S. R., and HESS, S. F. 1971. "Limit-Cycle Oscillations in Digital Filters," *IEEE Trans. Circuit Theory*, Vol. CT-18, pp. 687-696, November.
- PARKS, T. W., and McCLELLAN, J. H. 1972a. "Chebyshev-Approximation for Nonrecursive Digital Filters with Linear Phase," *IEEE Trans. Circuit Theory*, Vol. CT-19, pp. 189-194, March.
- PARKS, T. W., and McCLELLAN, J. H. 1972b. "A Program for the Design of Linear Phase Finite Impulse Response Digital Filters," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-20, pp. 195-199, August.
- PARZEN, E. 1957. "On Consistent Estimates of the Spectrum of a Stationary Time Series," *Am. Math. Stat.*, Vol. 28, pp. 329-348.
- PARZEN, E. 1974. "Some Recent Advances in Time Series Modeling," *IEEE Trans. Automatic Control*, Vol. AC-19, pp. 723-730, December.
- PEACOCK, K. L., and TREITEL, S. 1969. "Predictive Deconvolution-Theory and Practice," *Geophysics*, Vol. 34, pp. 155-169.
- PEEBLES, P. Z., JR. 1987. *Probability, Random Variables, and Random Signal Principles*, 2nd ed., McGraw-Hill, New York.
- PISARENKO, V. F. 1973. "The Retrieval of Harmonics from a Covariance Function," *Geophys. J. R. Astron. Soc.*, Vol. 33, pp. 347-366.
- POOLE, M. A. 1981. *Autoregressive Methods of Spectral Analysis*, E.E. degree thesis, Department of Electrical and Computer Engineering, Northeastern University, Boston, May.
- PRICE, R. 1990. "Mirror FFT and Phase FFT Algorithms," unpublished work, Raytheon Research Division, May.
- PROAKIS, J. G. 1995. *Digital Communications*, 3rd ed., McGraw-Hill, New York.
- RABINER, L. R., and SCHAEFER, R. W. 1974a. "On the Behavior of Minimax Relative Error FIR Digital Differentiators," *Bell Syst. Tech. J.*, Vol. 53, pp. 333-362, February.
- RABINER, L. R., and SCHAEFER, R. W. 1974b. "On the Behavior of Minimax FIR Digital Hilbert Transformers," *Bell Syst. Tech. J.*, Vol. 53, pp. 363-394, February.
- RABINER, L. R., and SCHAEFER, R. W. 1978. *Digital Processing of Speech Signals*, Prentice Hall, Englewood Cliffs, N.J.
- RABINER, L. R., SCHAEFER, R. W., and RADER, C. M. 1969. "The Chirp z -Transform Algorithm and Its Applications," *Bell Syst. Tech. J.*, Vol. 48, pp. 1249-1292, May-June.
- RABINER, L. R., GOLD, B., and MCGONEGAL, C. A. 1970. "An Approach to the Approximation Problem for Nonrecursive Digital Filters," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-18, pp. 83-106, June.

- RABINER, L. R., MCCLELLAN, J. H., and PARKS, T. W. 1975. "FIR Digital Filter Design Techniques Using Weighted Chebyshev Approximation," *Proc. IEEE*, Vol. 63, pp. 595-610, April.
- RADER, C. M. 1970. An Improved Algorithm for High-Speed Auto-correlation with Applications to Spectral Estimation," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-18, pp. 439-441, December.
- RADER, C. M., and BRENNER, N. M. 1976. "A New Principle for Fast Fourier Transformation," *IEEE Trans. Acoustics, Speech and Signal Processing*, Vol. ASSP-24, pp. 264-266, June.
- RADER, C. M., and GOLD, B. 1967a. "Digital Filter Design Techniques in the Frequency Domain," *Proc. IEEE*, Vol. 55, pp. 149-171, February.
- RADER, C. M., and GOLD, B. 1967b. "Effects of Parameter Quantization on the Poles of a Digital Filter," *Proc. IEEE*, Vol. 55, pp. 688-689, May.
- RAMSTAD, T. A. 1984. "Digital Methods for Conversion Between Arbitrary Sampling Frequencies," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-32, pp. 577-591, June.
- REMEZ, E. YA. 1957. *General Computational Methods of Chebyshev Approximation*, Atomic Energy Translation 4491, Kiev, USSR.
- RISSANEN, J. 1983. "A Universal Prior for the Integers and Estimation by Minimum Description Length," *Ann. Stat.*, Vol. 11, pp. 417-431.
- ROBERTS, R. A., and MULLIS, C. T. 1987. *Digital Signal Processing*, Addison-Wesley, Reading, Mass.
- ROBINSON, E. A. 1962. *Random Wavelets and Cybernetic Systems*, Charles Griffin, London.
- ROBINSON, E. A. 1982. "A Historical Perspective of Spectrum Estimation," *Proc. IEEE*, Vol. 70, pp. 885-907, September.
- ROBINSON, E. A., and TREITEL, S. 1978. "Digital Signal Processing in Geophysics," in *Applications of Digital Signal Processing*, A. V. Oppenheim, ed., Prentice Hall, Englewood Cliffs, N.J.
- ROBINSON, E. A., and TREITEL, S. 1980. *Geophysical Signal Analysis*, Prentice Hall, Englewood Cliffs, N.J.
- ROY, R., PAULRAJ, A., and KAILATH, T. 1986. "ESPRIT: A Subspace Rotation Approach to Estimation of Parameters of Cisoids in Noise," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-34, pp. 1340-1342, October.
- SAFRANEK, R. J., MACKEY, K. JAYANT, N. W., and KIM, T. 1988. "Image Coding Based on Selective Quantization of the Reconstruction Noise in the Dominant Sub-band," *Proc. 1988 IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 765-768, April.
- SAKAI, H. 1979. "Statistical Properties of AR Spectral Analysis," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-27, pp. 402-409, August.
- SANDBERG, I. W., and KAISER, J. F. 1972. "A Bound on Limit Cycles in Fixed-Point Implementations of Digital Filters," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-20, pp. 110-112, June.
- SATORIUS, E. H., and ALEXANDER J. T. 1978. "High Resolution Spectral Analysis of Sinusoids in Correlated Noise," *Proc. 1978 ICASSP*, pp. 349-351, Tulsa, Okla., April 10-12.

- SCHAFER, R. W., and RABINER, L. R. 1973. "A Digital Signal Processing Approach to Interpolation," *Proc. IEEE*, Vol. 61, pp. 692-702, June.
- SCHUEERMANN, H., and GOCKLER, H. 1981. "A Comprehensive Survey of Digital Transmultiplexing Methods," *Proc. IEEE*, Vol. 69, pp. 1419-1450.
- SCHMIDT, R. D. 1981. "A Signal Subspace Approach to Multiple Emitter Location and Spectral Estimation," Ph.D. dissertation, Department of Electrical Engineering, Stanford University, Stanford, Calif., November.
- SCHMIDT, R. D. 1986. "Multiple Emitter Location and Signal Parameter Estimation," *IEEE Trans. Antennas and Propagation*, Vol. AP 34, pp. 276-280, March.
- SCHOTT, J. P., and MCCLELLAN, J. H. 1984. "Maximum Entropy Power Spectrum Estimation with Uncertainty in Correlation Measurements," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-32, pp. 410-418, April.
- SCHÜR, I. 1917. "On Power Series Which Are Bounded in the Interior of the Unit Circle," *J. Reine Angew. Math.*, Vol. 147, pp. 205-232, Berlin. For an English translation of the paper, see Gohberg 1986.
- SCHUSTER, SIR ARTHUR. 1898. "On the Investigation of Hidden Periodicities with Application to a Supposed Twenty-Six-Day Period of Meteorological Phenomena," *Terr. Mag.*, Vol. 3, pp. 13-41, March.
- SEDLMEYER, A., and FETTWEIS, A. 1973. "Digital Filters with True Ladder Configuration," *Int. J. Circuit Theory Appl.*, Vol. 1, pp. 5-10, March.
- SHANKS, J. L. 1967. "Recursion Filters for Digital Processing," *Geophysics*, Vol. 32, pp. 33-51, February.
- SHANNON, C. E. 1949. "Communication in the Presence of Noise," *Proc. IRE*, pp. 10-21, January.
- SHEINGOLD, D. H., ed. 1986. *Analog-Digital Conversion Handbook*, Prentice Hall, Englewood Cliffs, N.J.
- SIEBERT, W. M. 1986. *Circuits, Signals and Systems*, McGraw-Hill, New York.
- SINGLETON, R. C. 1967. "A Method for Computing the Fast Fourier Transform with Auxiliary Memory and Limit High speed Storage," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-15, pp. 91-98, June.
- SINGLETON, R. C. 1969. "An Algorithm for Computing the Mixed Radix Fast Fourier Transform," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-17, pp. 93-103, June.
- SMITH, M. J. T., and BARWELL, T. P. 1984. "A Procedure for Designing Exact Reconstruction Filter Banks for Tree Structured Subband Coders," *Proc. 1984 IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 27.1.1-27.1.4, San Diego, March.
- SMITH, M. J. T., and EDDINS, S. L. 1988. "Subband Coding of Images with Octave Band Tree Structures," *Proc. 1987 IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 1382-1385, Dallas, April.
- STEIGLITZ, K. 1965. "The Equivalence of Digital and Analog Signal Processing," *Inf. Control*, Vol. 8, pp. 455-467, October.
- STEIGLITZ, K. 1970. "Computer-Aided Design of Recursive Digital Filters," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-18, pp. 123-129, June.

- STOCKHAM, T. G. 1966. "High Speed Convolution and Correlation," *1966 Spring Joint Computer Conference, AFIPS Proc.*, Vol. 28, pp. 229-233.
- STORER, J. E. 1957. *Passive Network Synthesis*, McGraw-Hill, New York.
- SWARZTRAUBER, P. 1986. "Symmetric FFT's," *Mathematics of Computation*, Vol. 47, pp. 323-346, July.
- SWINGLER, D. N. 1979a. "A Comparison Between Burg's Maximum Entropy Method and a Nonrecursive Technique for the Spectral Analysis of Deterministic Signals," *J. Geophys. Res.*, Vol. 84, pp. 679-685, February.
- SWINGLER, D. N. 1979b. "A Modified Burg Algorithm for Maximum Entropy Spectral Analysis," *Proc. IEEE*, Vol. 67, pp. 1368-1369, September.
- SWINGLER, D. N. 1980. "Frequency Errors in MEM Processing," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-28, pp. 257-259, April.
- SZEGÖ, G. 1967. *Orthogonal Polynomials*, 3rd ed., Colloquium Publishers, no. 23, American Mathematical Society, Providence, R.I.
- THORVALDSEN, T. 1981. "A Comparison of the Least-Squares Method and the Burg Method for Autoregressive Spectral Analysis," *IEEE Trans. Antennas and Propagation*, Vol. AP-29, pp. 675-679, July.
- TONG, H. 1975. "Autoregressive Model Fitting with Noisy Data by Akaike's Information Criterion," *IEEE Trans. Information Theory*, Vol. IT-21, pp. 476-480, July.
- TONG, H. 1977. "More on Autoregressive Model Fitting with Noise Data by Akaike's Information Criterion," *IEEE Trans. Information Theory*, Vol. IT-23, pp. 409-410, May.
- TRETTET, S. A. 1976. *Introduction to Discrete-Time Signal Processing*, Wiley, New York.
- TUFTS, D. W., and KUMARESAN, R. 1982. "Estimation of Frequencies of Multiple Sinusoids: Making Linear Prediction Perform Like Maximum Likelihood," *Proc. IEEE*, Vol. 70, pp. 975-989, September.
- ULRYCH, T. J., and BISHOP, T. N. 1975. "Maximum Entropy Spectral Analysis and Autoregressive Decomposition," *Rev. Geophys. Space Phys.*, Vol. 13, pp. 183-200, February.
- ULRYCH, T. J., and CLAYTON, R. W. 1976. "Time Series Modeling and Maximum Entropy," *Phys. Earth Planet. Inter.*, Vol. 12, pp. 188-200, August.
- VAIDYANATHAN, P. P. 1990. "Multirate Digital Filters, Filter Banks, Polyphase Networks, and Applications: A Tutorial," *Proc. IEEE*, Vol. 78, pp. 56-93, January.
- VAIDYANATHAN, P. P. 1993. *Multirate Systems and Filter Banks*, Prentice Hall, Englewood Cliffs, N.J.
- VETTERLI, J. 1984. "Multi-dimensional Sub-band Coding: Some Theory and Algorithms," *Signal Processing*, Vol. 6, pp. 97-112, April.
- VETTERLI, J. 1987. "A Theory of Multirate Filter Banks," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-35, pp. 356-372, March.
- VIEIRA, A. C. G. 1977. "Matrix Orthogonal Polynomials with Applications to Autoregressive Modeling and Ladder Forms," Ph.D. dissertation, Department of Electrical Engineering, Stanford University, Stanford, Calif., December.
- WALKER, G. 1931. "On Periodicity in Series of Related Terms," *Proc. R. Soc., Ser. A*, Vol. 313, pp. 518-532.
- WAX, M., and KAILATH, T. 1985. "Detection of Signals by Information Theoretic Criteria," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-32, pp. 387-392, April.

- WEINBERG, L. 1962. *Network Analysis and Synthesis*, McGraw-Hill, New York.
- WELCH, P. D. 1967. "The Use of Fast Fourier Transform for the Estimation of Power Spectra: A Method Based on Time Averaging over Short Modified Periodograms," *IEEE Trans. Audio and Electroacoustics*, Vol. AU-15, pp. 70-73, June.
- WIENER, N. 1949. *Extrapolation, Interpolation and Smoothing of Stationary Time Series with Engineering Applications*, Wiley, New York.
- WIENER, N., and PALEY, R. E. A. C. 1934. *Fourier Transforms in the Complex Domain*. American Mathematical Society, Providence, R.I.
- WINOGRAD, S. 1976. "On Computing the Direte Fourier Transform," *Proc. Natl. Acad. Sci.*, Vol. 73, pp. 105-106.
- WINOGRAD, S. 1978. "On Computing the Discrete Fourier Transform," *Math. Comp.*, Vol. 32, pp. 177-199.
- WOLD, H. 1938. *A Study in the Analysis of Stationary Time Series*, reprinted by Almqvist & Wiksell, Stockholm, 1954.
- WOOD, L. C., and TREITEL, S. 1975. "Seismic Signal Processing," *Proc. IEEE*, Vol. 63, pp. 649-661, April.
- WOODS, J. W., and O'NEIL, S. D. 1986. "Subband Coding of Images," *IEEE Trans. Acoustics, Speech, and Signal Processing*, Vol. ASSP-34, pp. 1278-1288, October.
- YOULA, D., and KAZANJIAN, N. 1978. "Bauer-Type Factorization of Positive Matrices and the Theory of Matrices Orthogonal on the Unit Circle," *IEEE Trans. Circuits and Systems*, Vol. CAS-25, pp. 57-69, January.
- YULE, G. U. 1927. "On a Method of Investigating Periodicities in Disturbed Series with Special References to Wolfer's Sunspot Numbers," *Philos. Trans. R. Soc. London, Ser. A*. Vol. 226, pp. 267-298, July.
- ZADEH, L. A., and DESOER, C. A. 1963. *Linear System Theory: The State-Space Approach*, McGraw-Hill, New York.
- ZVEREV, A. I. 1967. *Handbook of Filter Synthesis*, Wiley, New York.



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